Lana Nakai

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Education

University of British Columbia — Vancouver, BC
Bachelor of Science in Computer Science (Co-op) | GPA: 92.1% — *Sep. 2025 – May 2030 (expected)*

Technical Skills

Languages: C++, Python, Java, TypeScript, JavaScript, C, R, Racket, Bash, HTML/CSS
Frameworks & Libraries: ROS2, PX4, React.js, Next.js, Spring Boot, Tailwind CSS, gtest, pytest, JUnit
Developer Tools: Git, Docker, SSH, CMake, Bazel, Maven, Vite, Vercel, VS Code, Cursor, Eclipse, Claude Code
Relevant Courses: Software Construction, Data Structures and Algorithms (C++), Intro to Computer Systems, Discrete Mathematics, Matrix Algebra, Multivariable Calculus, Physics (Mechanics), Intro to Data Science, Intro to Statistics

Technical Experience

Software Developer — Sep. 2026 – Present
UBC UAS | Python, ROS2, WebRTC, Socket.IO, Docker, Git — *Vancouver, BC, Canada*

- Collaborate with a team of ∼20 software engineers to build end-to-end autonomy for a competition drone
- Rewrote the drone's onboard **WebRTC** signaling server to stream live video to multiple ground station frontends at once

Software Engineering Intern — Jul. 2026 – Sep. 2026
CHAOS Industries | C++, Python, Bash, Docker, Git — *San Francisco, CA, USA*

- Validated an in-house **Rust/C++ robotics framework** by applying it to a real-world use case (the Atropos data logger)
- Benchmarked the framework's runtime and memory usage with **Bash scripts** to identify performance bottlenecks
- Debugged across the company monorepo in close collaboration with mentors, following Git best practices

Software Engineering Intern — Jun. 2026 – Jul. 2026
Atropos Group | C++, Python, ROS2, SSH, Docker, Git — *Sunnyvale, CA, USA*

- Developed, tested (SIL/HIL), and deployed data logging software from the ground control station to the autopilot computer
- Built telemetry streaming over **UDP** for **real-time** flight data visualization
- Optimized verification logic to analyze **60,000+ messages** almost instantaneously
- Helped ensure a smooth technical transition following Atropos's acquisition by CHAOS Industries

Projects

Activity Generator | *Java, Spring Boot, JUnit, Maven, TypeScript, React.js, Next.js, Tailwind CSS* — Feb. 2026 – Present

- Designed and built a full-stack activity generator card game in Java, applying **OOP** principles to model decks and cards
- Developed a **RESTful API** with **Spring Boot** supporting CRUD operations on decks and cards, persisted via JPA and H2
- Migrated the frontend from a **Java Swing** GUI to a web app built with **TypeScript**, **React** (**Next.js**), and **Tailwind CSS**

Personal Website | *React.js, Tailwind CSS v4, JavaScript, Vite, Vercel* — May 2026

- Built and deployed a personal portfolio website on Vercel to showcase my projects
- Integrated the EmailJS API into a contact form, keeping API keys out of source control with environment variables

Motion | *HTML, CSS, JavaScript, Supabase, Git* — Apr. 2026

- Won **first place** in the youCode 2026 hackathon's beginner stream with an accessible web app tailored for non-profits
- Served as **full-stack developer**: built the frontend in HTML, CSS, and JavaScript and implemented user login with Supabase

Community Involvement

CPSC 110 Undergraduate Teaching Assistant — Jan. 2026 – Present
University of British Columbia — *Vancouver, BC*

- Support students in systematic program design and recursion in an introductory functional programming course
- **Co-lead** weekly labs of ∼30 students, explaining how to implement graph algorithms functionally and why they work
- Deliver lectures to roughly 60 students weekly, tailoring content to their needs